Theorem: Let (X, τ), (Y, ϑ) be topological spaces & $f: X \to Y$ be a function. Then f is τ-ϑ continuous $\iff$ $f(\bar{A}) \subseteq \overline{f(A)}$ for every subset A of X.

Proof:

$\Rightarrow$ Let $A \subseteq X$. Then $f(A) \subseteq Y$ and $\overline{f(A)} \in \vartheta_K$, since f is continuous $f^{-1}(\overline{f(A)}) \in \tau_K$. $\leftarrow$

$$A \subseteq f^{-1}(f(A)) \subseteq f^{-1}(\overline{f(A)})$$

$$\Rightarrow \bar{A} \subseteq \overline{f^{-1}(\overline{f(A)})} = f^{-1}(\overline{f(A)})$$

$$\Rightarrow f(\bar{A}) \subseteq \overline{f(A)}$$

$\Leftarrow$ Let $K \in \vartheta_K$, i.e. let K be a ϑ-closed subset of Y. Set $A = f^{-1}(K)$. We know from our hypothesis that $f(\bar{A}) \subseteq \overline{f(A)}$

or $\bar{A} \subseteq f^{-1}(\overline{f(A)})$.

$$f(\bar{A}) = f\left(\overline{f^{-1}(K)}\right) \subseteq \overline{f\left(f^{-1}(K)\right)} = \overline{f(A)}$$

$$\subseteq \bar{K} = K$$

$$\Rightarrow \overline{f^{-1}(K)} \subseteq f^{-1}(K)$$

$$\Rightarrow f^{-1}(K) \text{ is } \tau\text{-closed.}$$

$\therefore$ f is τ-ϑ continuous.

Defrnition: Let (X, τ) be a topological space & $A \subseteq X$.

(i) Let $x \in X$. If for every $U \in \tau$ with $x \in U$, $U \setminus \{x\} \cap A \neq \emptyset$ then x is called an accumulation point of A. The set of all accumulation points of A is called the derived set of A. It is denoted by $\tilde{A}$.

(ii) Let $x \in X$. If there exists a $U \in \tau$ s.t. $A \cap U = \{x\}$ then x is

called an isolated point of A.
In this course (although there is no such notation) I shall use the notation $I(A)$ in order to denote the set of all isolated points of A.

Note:

1. If $x \in \tilde{A}$ then x may or may not be an element of A.
2. If $x \in I(A)$ then $x \in A$ by definition. Thus $I(A) \subseteq A$.

Theorem: Let (X, τ) be a topological space & $A \subseteq X$.

(i) $\bar{A} = A \cup \tilde{A}$

(ii) $\bar{A} = \tilde{A} \cup I(A)$, where $\tilde{A} \cap I(A) = \phi$.

Proof:

(i) Let $x \in \overline{A}$. Since $A \subseteq \overline{A}$, x may or may not be in A.

Case 1 If $x \in A$ then $x \in A \cup \tilde{A}$

Case 2 Let $x \in \overline{A} \setminus A$. Because $x \in \overline{A}$, $A \cap U \neq \emptyset$ for every $U \in \tau$ with $x \in U$. Because $x \notin A$ & $A \cap U \neq \emptyset$, $A \cap U \setminus \{x\} \neq \emptyset$ for every $U \in \tau$ with $x \in U$. Then $x \in \tilde{A}$.

$\Rightarrow x \in A \cup \tilde{A}$

$\therefore \overline{A} \subseteq A \cup \tilde{A}$ ..... (1)

Let $x \in A \cup \tilde{A}$. Then $x \in A$ or $x \in \tilde{A}$

- If $x \in A$ then $x \in \overline{A}$.
- Assume that $x \in \tilde{A} \setminus A$. Because $x \in \tilde{A}$

$\emptyset \neq U \setminus \{x\} \cap A \subseteq U \cap A$

& hence $U \cap A \neq \emptyset$ for any $U \in \tau$ with $x \in U$. $\Rightarrow x \in \overline{A}$.

$\therefore A \cup \tilde{A} \subseteq \overline{A}$ ..... (2)

By (1) & (2) we have the equality.

(ii) Let $x \in \overline{A}$. Because $I(A) \subseteq A \subseteq \overline{A}$, x may or may not be in $I(A)$.

- If $x \in I(A)$ then $x \in I(A) \cup \tilde{A}$.
- Assume that $x \in \overline{A} \setminus I(A)$.

For every $U \in \tau$ with $x \in U$ $U \cap A \neq \emptyset$ since $x \in \overline{A}$. Moreover since $x \notin I(A)$, $\{x\} \subset U \cap A$ ($\neq$)

$\Rightarrow$ $U \setminus \{x\} \cap A \neq \emptyset$ for every $U \in \tau$ with $x \in U$.

$\Rightarrow$ $x \in \tilde{A}$

$\Rightarrow$ $x \in \tilde{A} \cup I(A)$

$\therefore$ $\overline{A} \subseteq \tilde{A} \cup I(A)$ ..... (1)

Conversely let $x \in \tilde{A} \cup I(A)$. Then $x \in I(A)$ or $x \in \tilde{A}$.

- If $x \in I(A)$, $x \in \overline{A}$
- Assume that $x \in \tilde{A}$ Then for every $U \in \tau$ with $x \in U$

$$\phi \neq U \setminus \{x\} \cap A \subseteq U \cap A$$

$$\Rightarrow x \in \overline{A}$$

$$\therefore \tilde{A} \cup I(A) \subseteq \overline{A} \quad \ldots \quad (2)$$

Now the equality follows from (1) & (2).

Suppose, to the contrary, that $I(A) \cap \tilde{A} \neq \phi$. Then $x \in I(A) \cap \tilde{A}$ for some $x \in X$.

$x \in I(A) \Rightarrow A \cap U = \{x\}$ for some $U \in \tau$.

$\Rightarrow U \setminus \{x\} \cap A = \phi$

$\Rightarrow x \notin \tilde{A}$. A contradiction.

$\therefore \tilde{A} \cap I(A) = \phi$.

Examples:

1. Let (X, τ) be an indiscrete topological space & $A \subseteq X$. Determine $\tilde{A}$ & $I(A)$.

Sol^

Case 1 Let $A = \{x\}$ for some $x \in X$.

Let us check if $x \in \tilde{A}$:

Since $\tau = \{\phi, X\}$, there is only one open set containing x namely X itself. Moreover

$$X \setminus \{x\} \cap A = X \setminus \{x\} \cap \{x\} = \phi$$

$\therefore x \notin \tilde{A}$.

Let $y \in X \setminus \{x\} = X \setminus A$. Again X is the only τ-open subset of X containing y &

$$X \setminus \{y\} \cap A = X \setminus \{y\} \cap \{x\} = \{x\} \neq \phi$$

$\Rightarrow$ If $y \in X \setminus \{x\}$ then $y \in \tilde{A}$.

$$\therefore \tilde{A} = X \setminus \{x\} = X \setminus A$$

There is only one open set containing x, namely X itself &

$$X \cap A = X \cap \{x\} = \{x\} \Rightarrow I(A) = \{x\} = A$$

OR

Any nonempty subset A of an indiscrete topological space is a dense subset of the topological space.

Thus $\overline{A} = \overline{\{x\}} = X$.

Moreover $\overline{A} = \tilde{A} \cup I(A)$ $\quad \tilde{A} = X \setminus A$

Thus $I(A) = \overline{A} \setminus \tilde{A}$

$= X \setminus (X \setminus A)$

$= A$.

Case 2 Assume that $A \neq \{x\}$ for any $x \in X$, i.e. assume that A is not a singleton.

$I(A) = ?$

The only nonempty open set is X.

$X \cap A = A \neq$ a singleton.

$\Rightarrow$ None of the pt's in A will be isolated points of A.

$\Rightarrow$ $I(A) = \emptyset$

Because $\overline{A} = X$ & $I(A) = \emptyset$

$$\tilde{A} = \overline{A} \setminus I(A) = X \setminus \emptyset = X.$$

∴ If we summarize we have:

$$\tilde{A} = \begin{cases} X & \text{if } A \text{ contains at least 2 points} \\ X \setminus \{x\} & \text{if } A = \{x\} \end{cases}$$

AND

$$I(A) = \begin{cases} \{x\} & \text{if } A = \{x\} \\ \emptyset & \text{if } A \text{ contains at least 2 pt's.} \end{cases}$$

2. Let (X, τ) be a discrete topological space & $A \subseteq X$. Determine $I(A)$ & $\tilde{A}$.

Solution:

Because $\{x\} \in \tau$ for each $x \in X$ and $\{x\} \setminus \{x\} \cap A = \emptyset$ for any $A \subseteq X$ $\tilde{A} = \emptyset$. Moreover because $\{x\} \in \tau$ & $\{x\} \cap A = \{x\}$ for every $x \in A$ $I(A) = A$.

Example: Let (X, τ) be a topological space & $A \subseteq X$. Does the equality $\tilde{A} = \tilde{\tilde{A}}$ always hold?

Sol^n No, not in general.

Consider the indiscrete topological space (X, τ) & $A = \{x\}$ for some $x \in X$. We know that $\overline{A} = X$, $\tilde{A} = X \setminus \{x\}$. But $\tilde{\tilde{A}} = \tilde{X} = X$, i.e. $\tilde{A} \subset \tilde{\tilde{A}}$.
$\neq$

Example: Let $X = \mathbb{Z}^+$ = the set of all positive integers & let

$\tau = \{ U \mid U \subseteq X$ and $(n$ is an odd integer & $n \in U) \Rightarrow n+1 \in U \}$.

(i) Show that τ is a topology on X.

(ii) Let $A = \{1, 2\}$, $B = \{1, 2, 3\}$, $C = \{2, 4, 5\}$ & $D = \{1, 2, 3, 5, 7, 9, \ldots, 2n+1, \ldots\}$.

Determine the interiors, exteriors, boundaries, closures, derived sets & sets of isolated points of A, B, C & D.

Sol^n

(i) Exercise.

(ii) $\mathring{A} = A = \{1, 2\}$

$X \setminus A = \mathbb{Z}^+ \setminus \{1, 2\} = \{3, 4, 5, 6, \ldots\}$

$\Rightarrow$ $ext(A) = X \setminus A = \{3, 4, 5, 6, \ldots\}$

$\partial A = X \setminus [\mathring{A} \cup ext(A)]$

$= \mathbb{Z}^+ \setminus \mathbb{Z}^+ = \phi$

$\overline{A} = \mathring{A} \cup \partial A$ or $\overline{A} = X \setminus ext(A)$

$\overline{A} = \{1,2\} = A$ or $\overline{A} = \mathbb{Z}^+ \setminus \{3,4,\ldots\}$

$= \{1,2\} = A$

$I(A) = \{2\}$:

If $1 \in I(A)$ then there would be a $U \in \mathcal{T}$ s.t. $U \cap A = \{1\}$. But any open set U containing must contain 2 as well. Hence $U \cap A \neq \{1\}$

$\therefore 1 \notin I(A)$.

Is $2 \in I(A)$? $\{2\} \in \mathcal{T}$ &

$\{2\} \cap A = \{2\} \Rightarrow 2 \in I(A)$

$\tilde{A} = ?$

$\overline{A} = \tilde{A} \dot{\cup} I(A)$. Here

$\overline{A} = A$, $I(A) = \{2\}$. Thus

$\tilde{A} = A \setminus I(A) = \{1,2\} \setminus \{2\} = \{1\}$.

$B = \{1, 2, 3\}$

$\mathring{B} = \{1, 2\}$.

$X \setminus B = \{4, 5, 6, \ldots\}$

$ext(B) = (X \setminus B)^\circ = X \setminus B = \{4, 5, 6, \ldots\}$.

$\partial B = X \setminus [\mathring{B} \cup ext(B)]$

$= \mathbb{Z}^+ \setminus (\mathbb{Z}^+ \setminus \{3\})$

$= \{3\}$

$\overline{B} = B \cup \partial B = \{1, 2, 3\} = B$.

$I(B) = \{2, 3\}$

$1 \notin I(B)$

$\{2\} \in \tau$, $\{2\} \cap B = \{2\} \Rightarrow 2 \in I(B)$

$3 \in \{3, 4\} \in \tau$, $\{3, 4\} \cap B = \{3\} \Rightarrow 3 \in I(B)$

$\tilde{B} = \overline{B} \setminus I(B) = \{1, 2, 3\} \setminus \{2, 3\}$

$= \{1\}$

$C = \{2, 4, 5\}$

$\mathring{C} = \{2, 4\}$

$X \setminus C = \{1, 3, 6, 7, \ldots\}$

$ext(C) = (X \setminus C)^\circ = \{6, 7, 8, \ldots\}$

$\partial C = X \setminus [\mathring{C} \cup ext(C)]$

$= \{1, 3, 5\}$

$\overline{C} = C \cup \partial C = \{1, 2, 3, 4, 5\}$

$I(C) = \{2, 4, 5\} = C$

$\tilde{C} = \overline{C} \setminus I(C) = \{1, 2, 3, 4, 5\} \setminus \{2, 4, 5\}$

$= \{1, 3\}$

$D = \{1, 2, 3, 5, 7, 9, \ldots\}$

$\mathring{D} = \{1, 2\}$

$X \setminus D = \{4, 6, 8, \ldots\}$

$ext(D) = (X \setminus D)^\circ = \{4, 6, 8, \ldots\}$

$= X \setminus D$

$\partial D = X \setminus \{1, 2, 4, 6, 8, 10, \ldots\}$

$= \{3, 5, 7, 9, \ldots\}$

$$\overline{D} = \{1, 2, 3, 5, 7, \ldots\} = D$$

$$I(D) = \{2, 3, 5, 7, 9, \ldots\} = D \setminus \{1\}$$

$$\tilde{D} = \overline{D} \setminus I(D) = D \setminus (D \setminus \{1\}) = \{1\}$$

Neighbourhoods & neighbourhood systems

Definition: Let (X, τ) be a topological space, $A, N \subseteq X$. If there exists a $U \in \tau$ s.t. $A \subseteq U \subseteq N$ then N is called a neighbourhood of A.

The class consisting of all neighbourhoods of A is denoted by $\eta(A)$ & it is called the neighbourhood system of A, i.e.

$\eta(A) = \{N \mid N \subseteq X$ and $A \subseteq U \subseteq N$ for some $U \in \tau\}$ is the neighbourhood system of A.

In particular if $A = \{x\}$ then